

Computer Networks (CL3001)



CN LAB 09

Submitted by:

Syeda Sara Ali

23K-0070

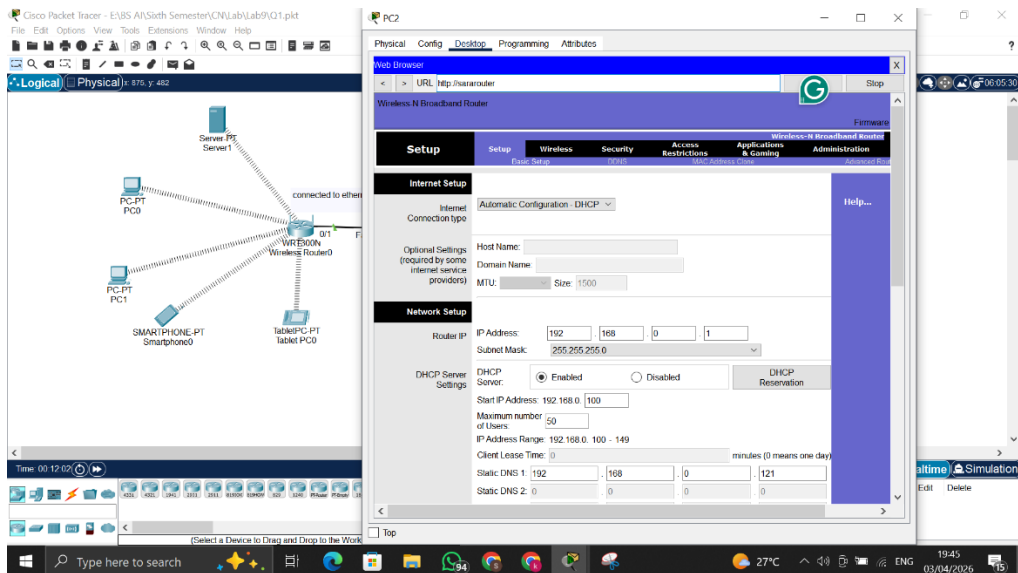
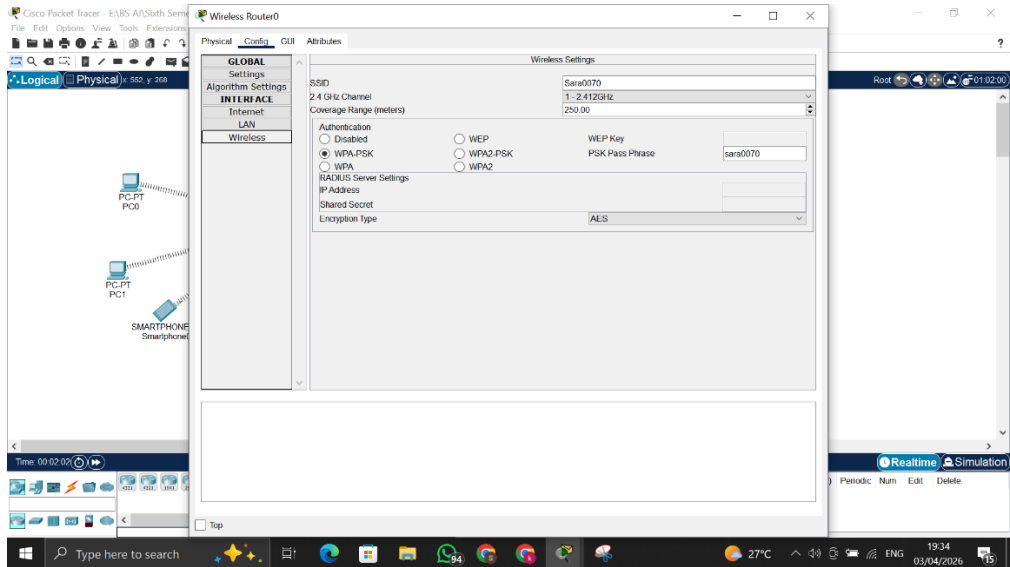
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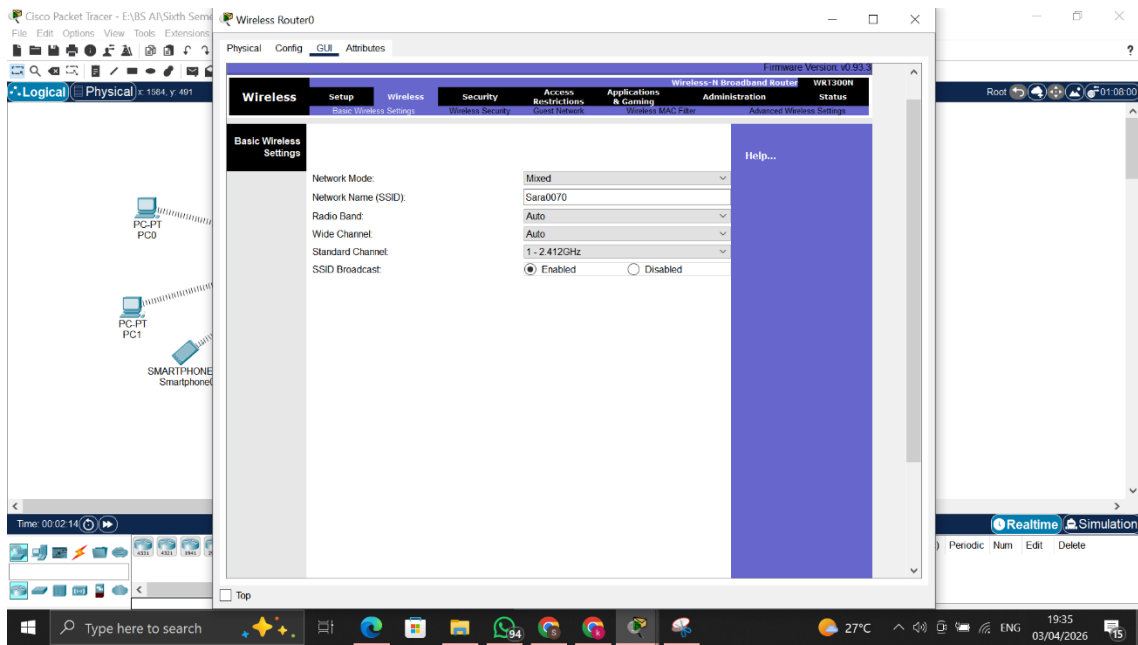
Submitted to:

Sir Sameer Faisal

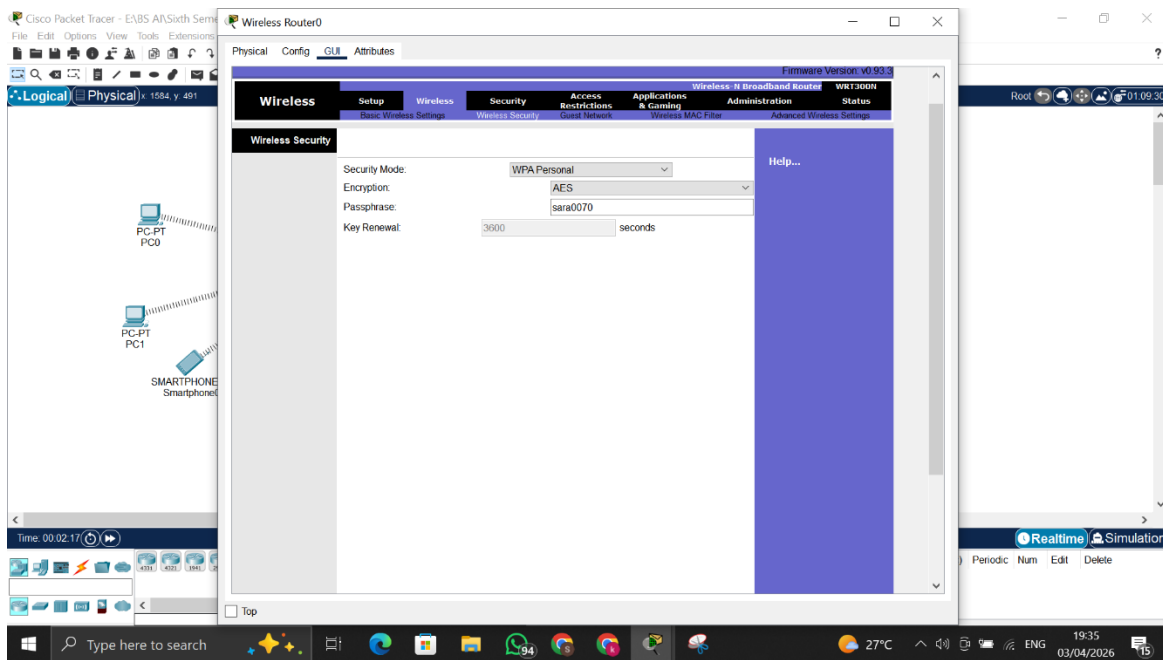
Lab Tasks 1:

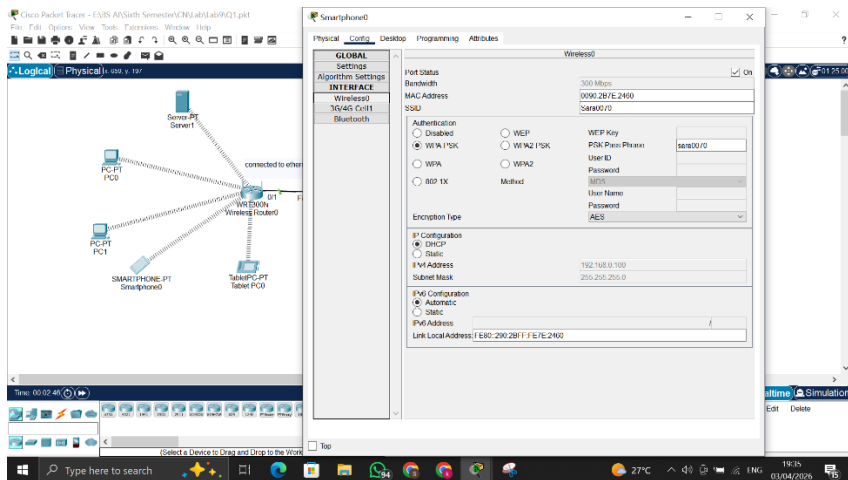
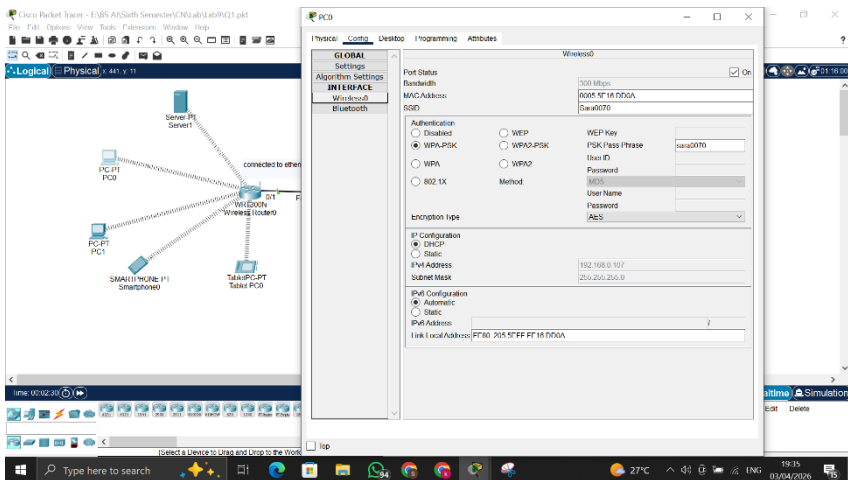
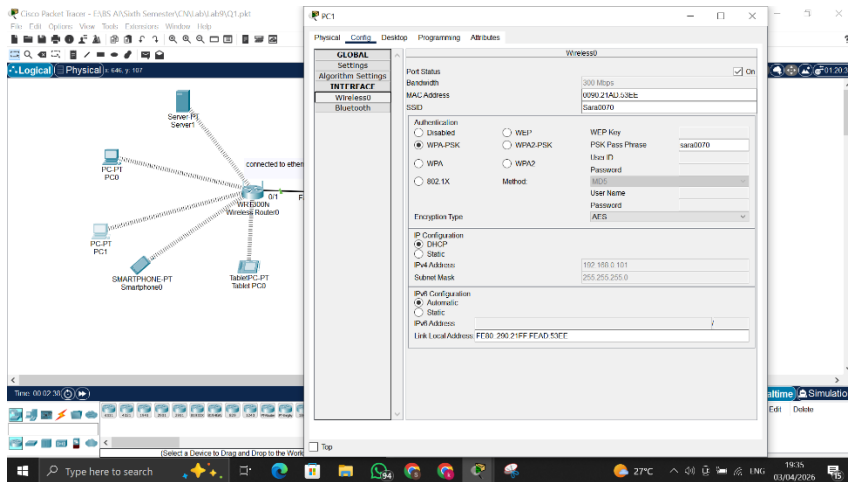
1. *Change the Network of the Wireless Router to your StudentID+Name.*

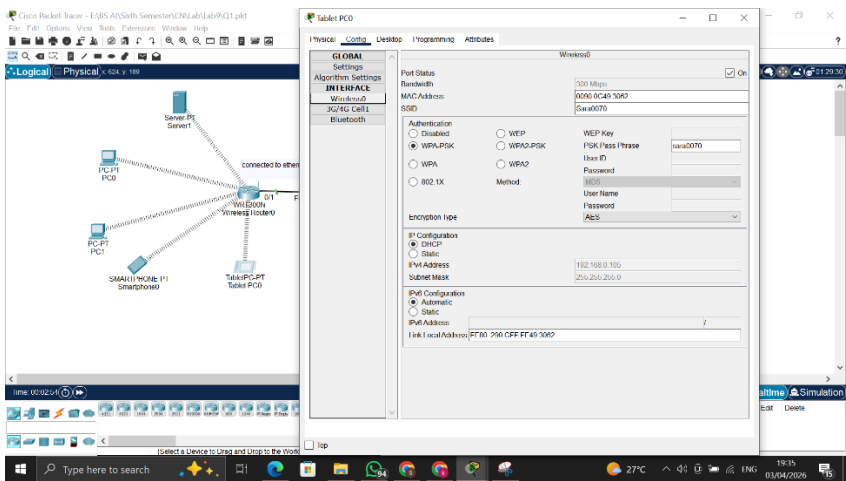
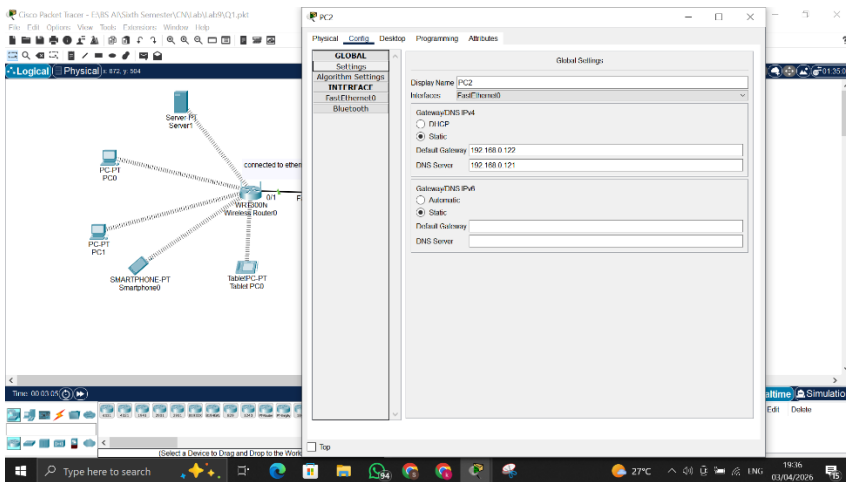
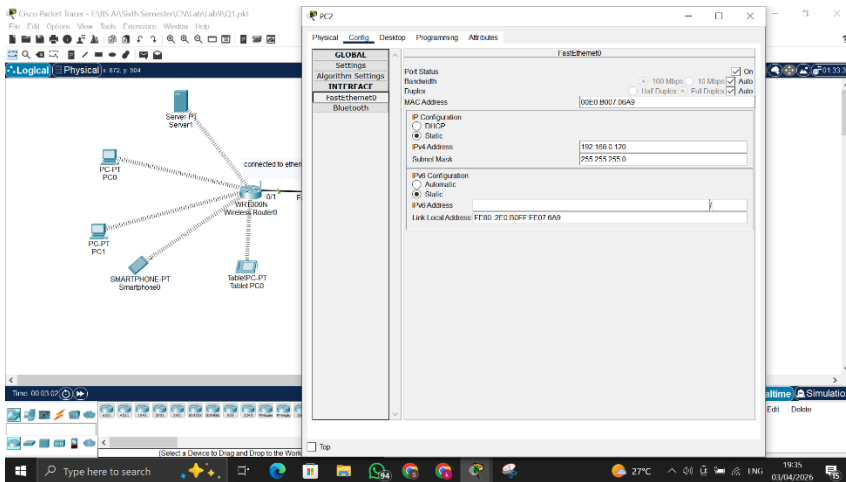


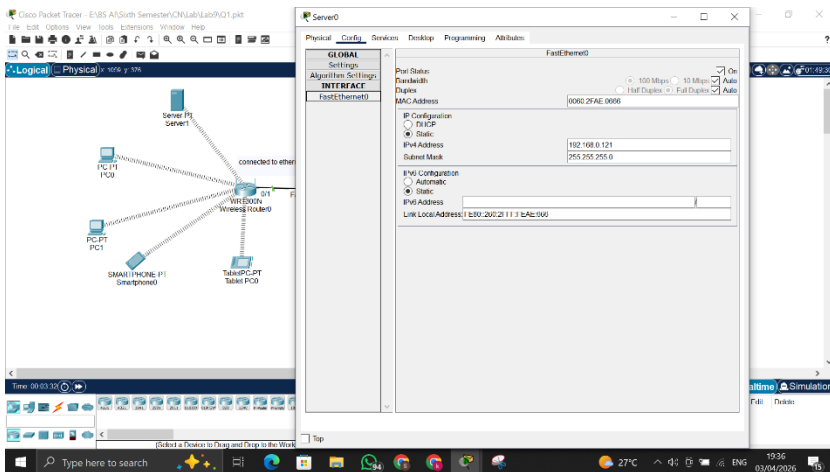
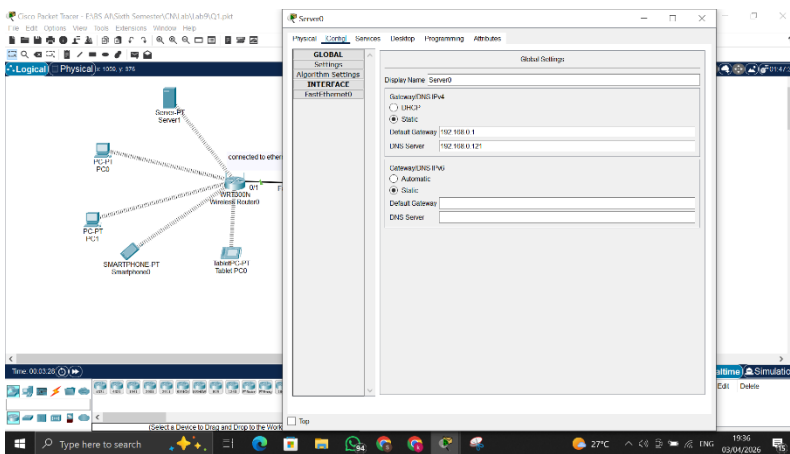
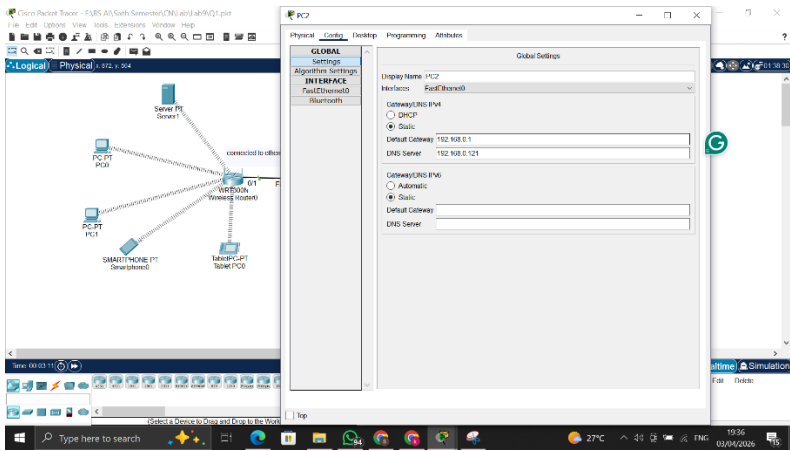


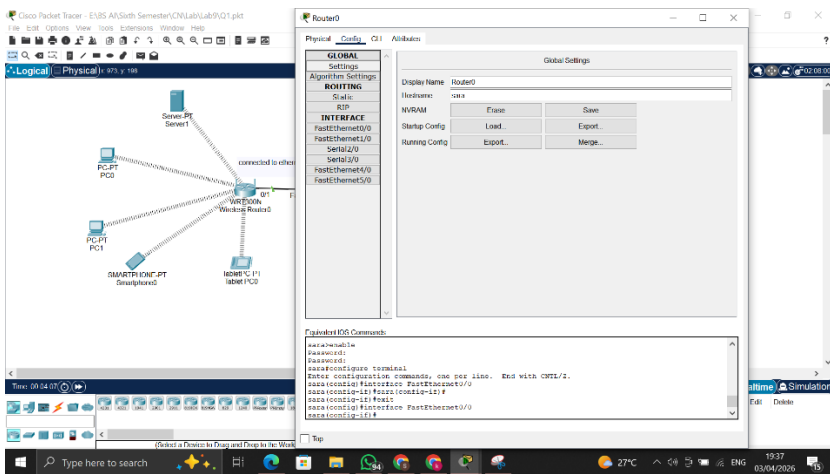
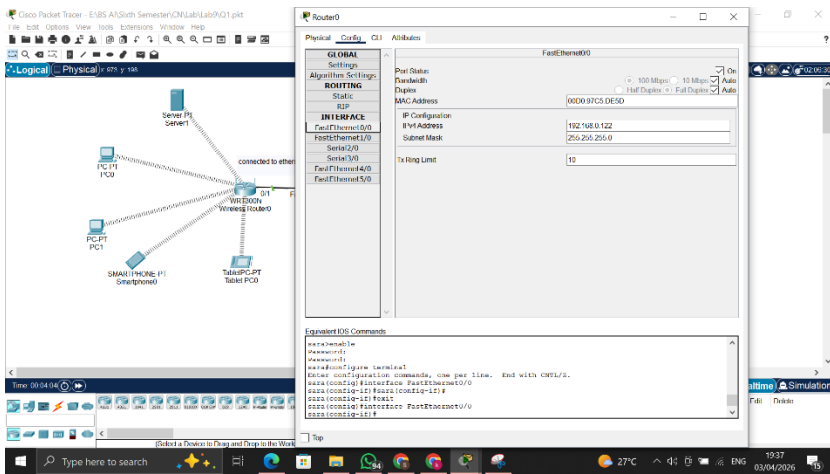
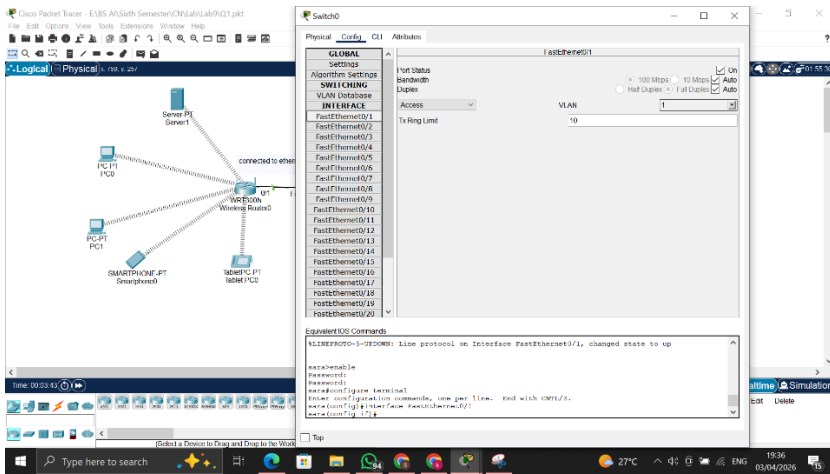
2. Set the key while connecting the wireless router with end devices.

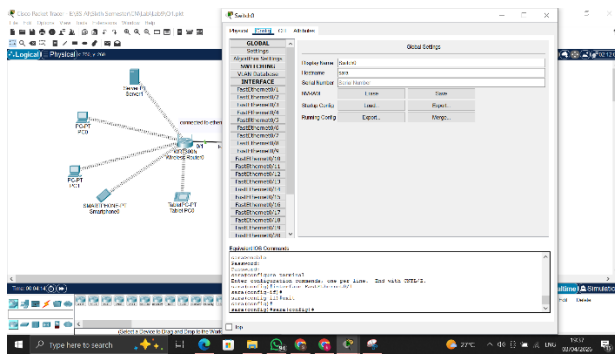




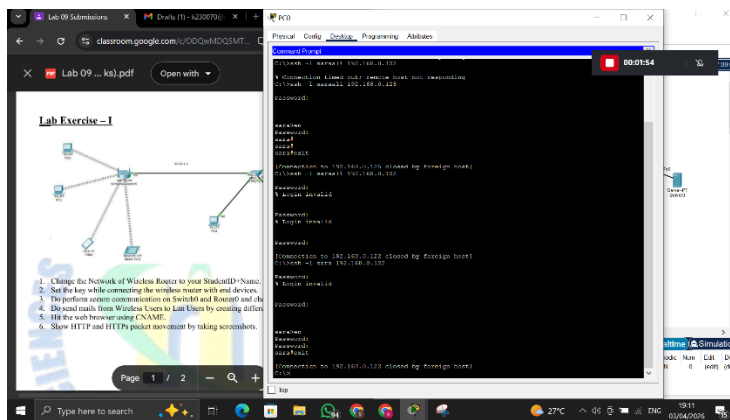




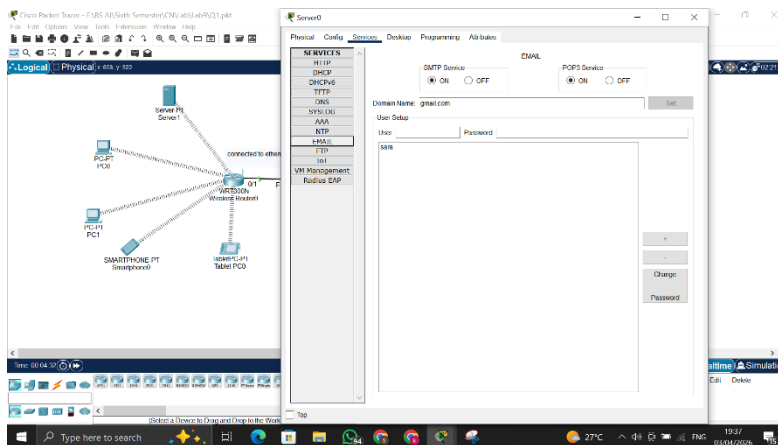


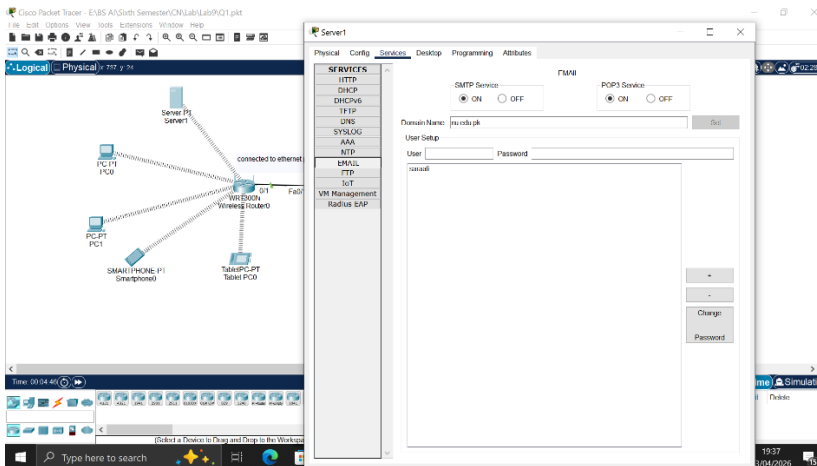
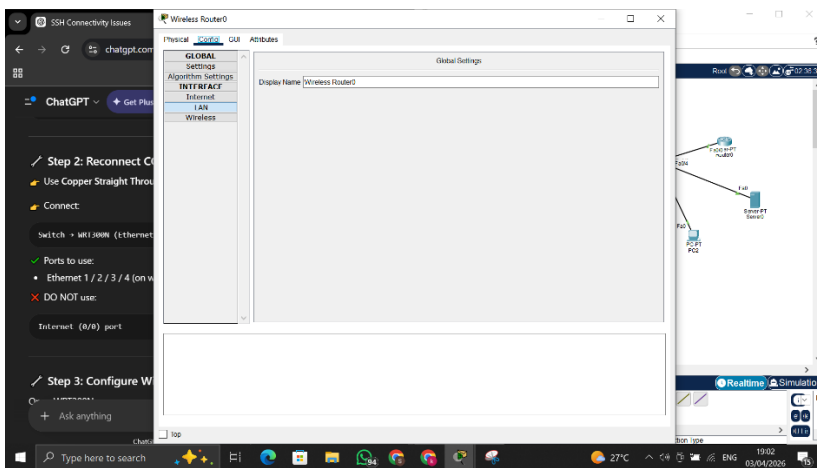
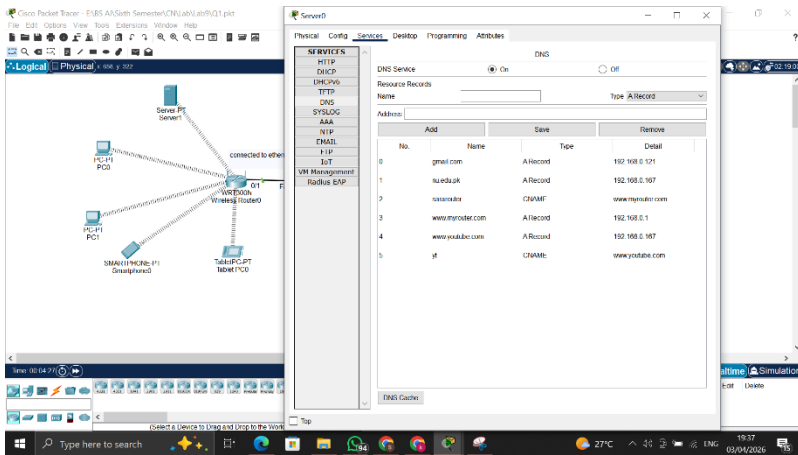


3. Do perform secure communication on Switch0 and Router0 and check it through PC0.



4. Do send mails from Wireless Users to Lan Users by creating different domains.





The network topology shows a central 'Wireless Router' connected to a 'Server-PT Server1', two 'PC-PT' devices, and two mobile devices: 'SMARTPHONE-PT Smartphone0' and 'Tablet-PT Tablet PC0'. The mail browser window on the smartphone displays the following email list:

	From	Subject	Received
1	server@gmail.com	dmad	Fri Apr 3 2020 19:20:29
2	Mail Delivery Subsystem	Delivery Status Notification (failure)	Fri Apr 3 2020 19:16:57

The email content for the first entry is:

dmad
server@gmail.com
Sent: Fri Apr 3 2020 19:20:29
checkmate wc

The network topology is identical to the first screenshot. The mail browser window on the PC displays the following email list:

	From	Subject	Received
1	server@nu.edu.pk	hi from nu	Fri Apr 3 2020 19:16:52

The email content for the first entry is:

hi from nu
server@nu.edu.pk
Sent: Fri Apr 3 2020 19:16:52
hello dars

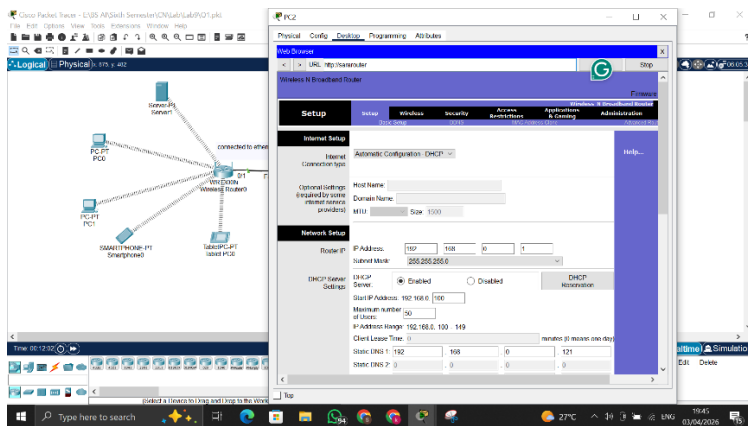
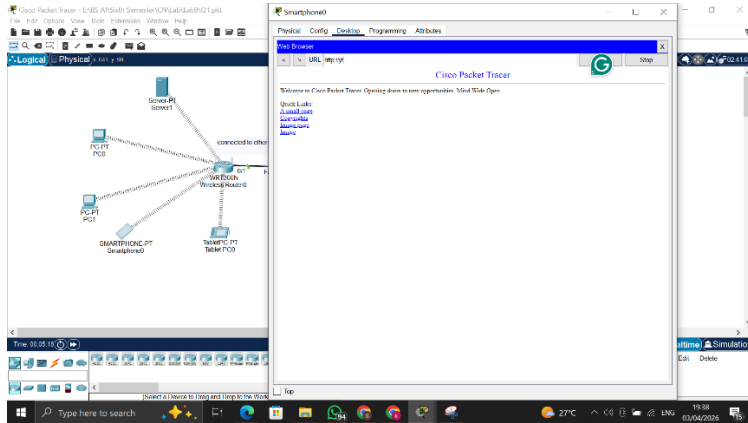
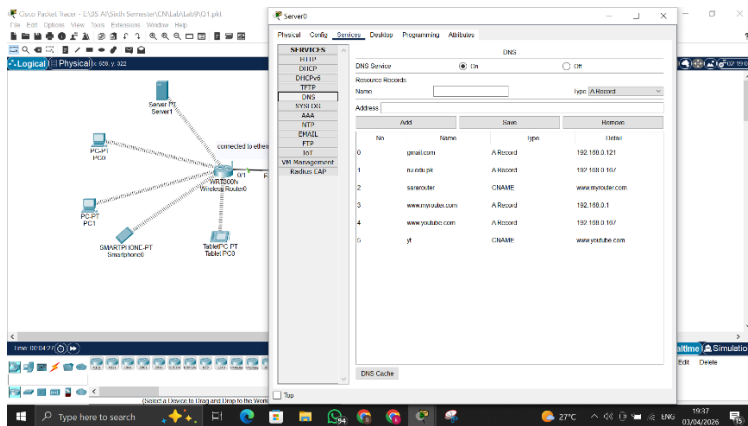
The network topology is identical to the previous screenshots. The mail browser window on the smartphone displays the following email list:

	From	Subject	Received
1	server@gmail.com	dmad	Fri Apr 3 2020 19:20:29
2	Mail Delivery Subsystem	Delivery Status Notification (failure)	Fri Apr 3 2020 19:16:57

The email content for the first entry is:

dmad
server@gmail.com
Sent: Fri Apr 3 2020 19:20:29
checkmate wc

5. Hit the web browser using CNAME.



6. Show HTTP and HTTPs packet movement by taking screenshots

The screenshot shows the Cisco Packet Tracer interface with a network topology and the Event List panel. The network includes a central switch (S1) connected to various devices: Server-PT, PC-PT, Smartphone-PT, Tablet-PT, and several wireless routers. The Event List panel displays a list of HTTPs events.

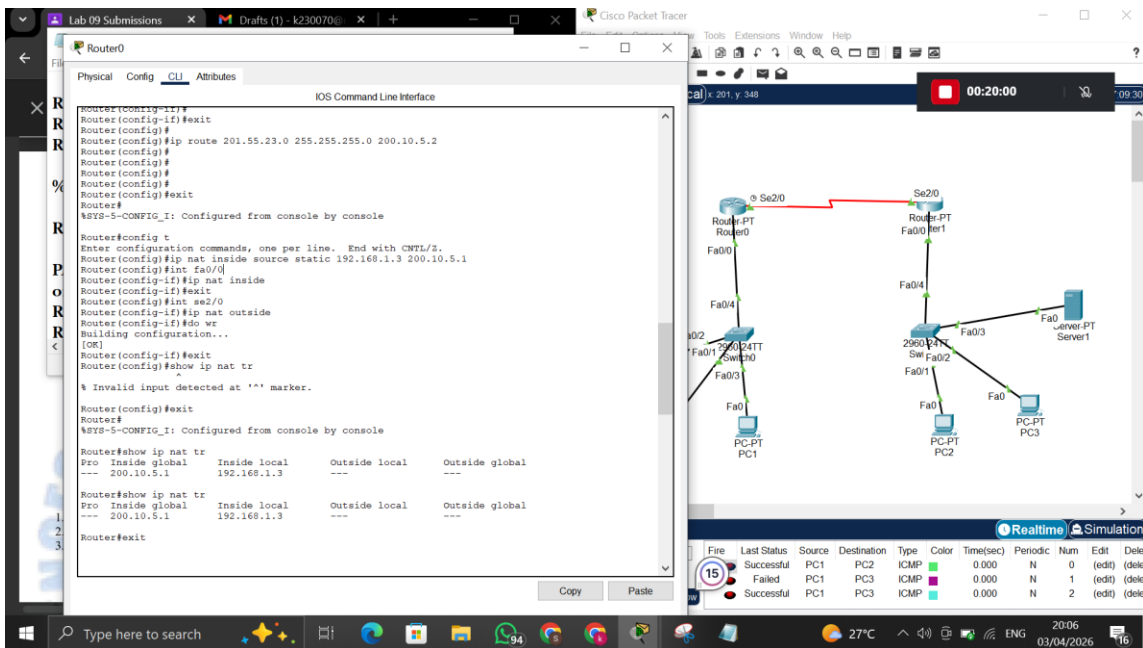
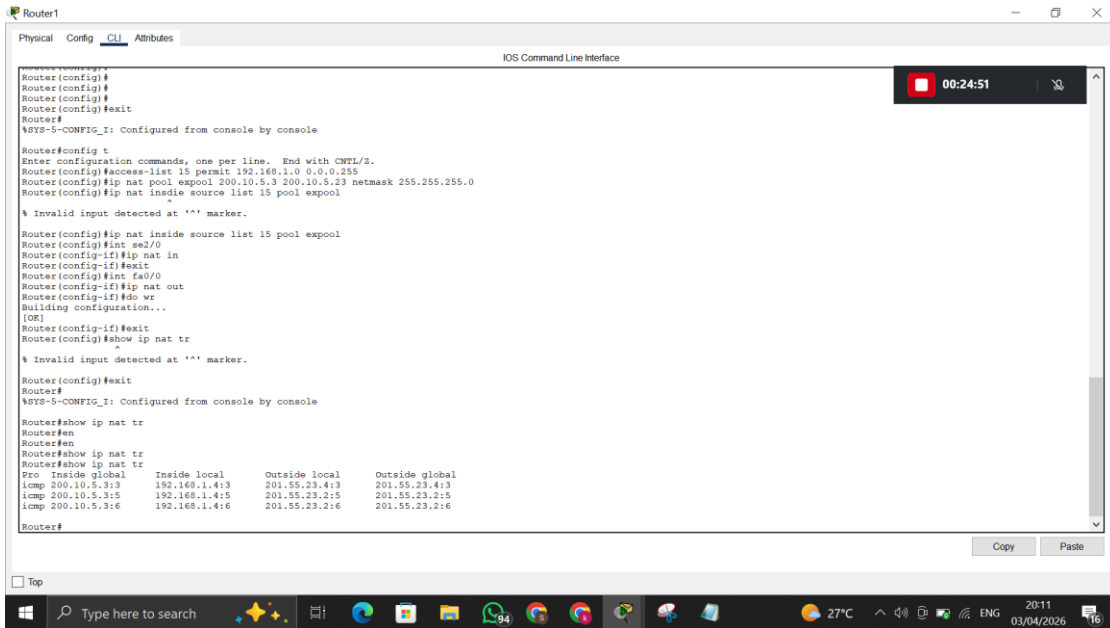
Vis.	Time(sec)	Last Device	At Device	Type
0.253	—	—	PC1	HTTPs
0.254	—	PC1	Wireless Router0	HTTPs
0.276	—	—	Wireless Router0	HTTPs
0.277	—	Wireless Router0	PC0	HTTPs
0.277	—	Wireless Router0	PC1	HTTPs
0.277	—	Wireless Router0	Tablet PC0	HTTPs
0.277	—	Wireless Router0	Smartphone0	HTTPs
0.277	—	Wireless Router0	Server1	HTTPs
0.278	—	—	Server1	HTTPs
0.279	—	Server1	Wireless Router0	HTTPs
0.280	—	—	Wireless Router0	HTTPs
Visible 0.281	—	Wireless Router0	PC0	HTTPs
Visible 0.281	—	Wireless Router0	PC1	HTTPs
Visible 0.281	—	Wireless Router0	Smartphone0	HTTPs
Visible 0.281	—	Wireless Router0	Tablet PC0	HTTPs
Visible 0.281	—	Wireless Router0	Server1	HTTPs

The screenshot shows the Cisco Packet Tracer interface with the same network topology as the first screenshot. The Event List panel displays a list of HTTP events.

Vis.	Time(sec)	Last Device	At Device	Type
450.089	—	PC1	Wireless Router0	HTTP
450.099	—	—	Wireless Router0	HTTP
450.100	—	Wireless Router0	PC0	HTTP
450.100	—	Wireless Router0	PC1	HTTP
450.100	—	Wireless Router0	Tablet PC0	HTTP
450.100	—	Wireless Router0	Smartphone0	HTTP
450.100	—	Wireless Router0	Server1	HTTP
450.109	—	—	Server1	HTTP
450.110	—	Server1	Wireless Router0	HTTP
450.118	—	—	Wireless Router0	HTTP
450.119	—	Wireless Router0	PC0	HTTP
450.119	—	Wireless Router0	PC1	HTTP
450.119	—	Wireless Router0	Tablet PC0	HTTP
450.119	—	Wireless Router0	Smartphone0	HTTP
450.119	—	Wireless Router0	Server1	HTTP
Visible 450.138	—	—	PC1	HTTP

Lab Tasks 2:

1. Perform Static NAT on Router0 and Dynamic NAT on Router1.



Router1

Physical Config Attributes

Static Routes

Network Address: 192.168.1.0/24 via 200.10.5.1

Equivalent IOS Commands

```

Router(config)# interface FastEthernet0/0
Router(config)# ip address 200.10.5.1 255.255.255.0
Router(config)# ip route 192.168.1.0 255.255.255.0 200.10.5.1
Router(config)#

```

Lab Exercise - II

1. Perform Static Nat on Router0 and Dynamic Nat on Router1.
2. Do send mail from PC1 to PC2.
3. Do hit the website from PC0.

Page 2 / 2

Router0

Physical Config CU Attributes

Static Routes

Network Address: 201.55.23.0/24 via 200.10.5.1

Equivalent IOS Commands

```

Router(config)# interface Serial3/0
Router(config)# ip address 200.10.5.1 255.255.255.0
Router(config)# ip route 201.55.23.0 255.255.255.0 200.10.5.1
Router(config)#

```

Lab Exercise - II

1. Perform Static Nat on Router0 and Dynamic Nat on Router1.
2. Do send mail from PC1 to PC2.
3. Do hit the website from PC0.

Page 2 / 2

Cisco Packet Tracer - EIS5 AI/Sixth Semester/CN/Lab/Lab09/02.pkt

Logical Physical

Time: 00:04:11

Realtime Simulation

Scenario 0

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
		Mail	PC2	ICMP		0.000	N	0	(edit)	(delete)
		PC2	PC1	ICMP		0.000	N	1	(edit)	(delete)
		PC2	Mail	ICMP		0.000	N	2	(edit)	(delete)

2. Do send mail from PC1 to PC2.

The network diagram shows a topology with two switches (S200, S201) connected to two routers (R1, R2). PC1 is connected to R1, and PC2 is connected to R2. A mail server (Mail) is connected to R1. The PC1 Mail Browser interface is shown, displaying the 'Compose' button and the 'Send' button. The status bar indicates 'Sending mail to sakina@gmail.com, with subject: dng d.. Mail Server: 192.168.1.2 Send Success.' The simulation time is 00:17:00.

The network diagram is the same as the previous one. The PC2 Mail Browser interface is shown, displaying the 'Receive' button. The status bar indicates 'Receiving mail from POP3 Server 192.168.1.2 Receive Mail Success.' The simulation time is 00:18:51.

	From	Subject	Received
1	sara@gmail.com	dng d	Fri Apr 3 2020 19:52:33
2	sara@gmail.com	hiiii	Fri Apr 3 2020 21:13:37
3	sara@gmail.com	hiiii	Fri Apr 3 2020 20:53:17

classroom.google.com/c/ODQwMQQ5MT...

Lab Exercise - II

1. Perform Static Nat on Router1 and Dynamic Nat on Router1.
2. Do send mail from PC1 to PC2.
3. Do hit the website from PC2.

Page 2 / 2

Server0

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- FTP
- DHCPv6
- TFTP
- DNS
- SYNLOG
- AAA
- NTP
- (Dial)
- IPsec
- ICMP
- VM Management
- Radius RADIUS

DNS Service

DNS Service: ☒ On ☐ Off

Resource Records

Name: Type:

Host Name

No.	Name	Type	Detail
0	www.nxt.com	A Record	192.168.1.2
1	nxt	CNAME	www.nxt.com

DNS Cache

Lab Ex

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Display Name: PC0

Interfaces: FastEthernet0

Gateway (DNS v4)

☐ DHCP

☒ Static

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.2

Gateway (DNS v6)

☐ Automatic

☒ Static

Default Gateway:

DNS Server:

Server1

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYNLOG
- AAA
- NTP
- (Dial)
- IPsec
- ICMP
- VM Management
- Radius RADIUS

DNS Service

DNS Service: ☒ On ☐ Off

Resource Records

Name: Type:

Address

No.	Name	Type	Detail
0	www.nxt.com	A Record	192.168.1.2
1	nxt	CNAME	www.nxt.com

DNS Cache

PC1

Physical Config Desktop Programming Attributes

Configure Mail

User Information

Your Name: user

Email Address: user@gmail.com

Server Information

Incoming Mail Server: 192.168.1.2

Outgoing Mail Server: 192.168.1.2

Login Information

User Name: user

Password: ****

Save Remove Clear Reset

Mail Server

Cancel Send/Receive

3. Do hit the website from PC0

The screenshot displays the Cisco Packet Tracer interface. On the left, a PC named PC0 is shown with a web browser window open. The browser's address bar contains the URL `http://xyz`. The page content includes a welcome message and quick links: [A small page](#), [Copies](#), [Images page](#), and [Image](#). Below the browser, a list of tasks is visible: 1. Perform, 2. Do send, 3. Do hit t.

On the right, the network topology is shown. It consists of two routers (Router-PT) connected via their Serial (Se2/0) ports. The left router is connected to a switch (2960-PT) via its Fa0/4 port. The switch is connected to PC1 via its Fa0/3 port. The right router is connected to a switch (2960-PT) via its Fa0/4 port. This switch is connected to PC2 via its Fa0/1 port and to PC3 via its Fa0/2 port. A server (Server-PT) is also connected to the right switch via its Fa0/3 port. The status bar at the bottom indicates that the simulation is running in Realtime mode. A table shows the results of the simulation:

re	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delc
●	Successful	PC2	PC1	ICMP	0.000	N	2	(edit)	(delc)	
●	Successful	Server0	PC2	ICMP	0.000	N	3	(edit)	(delc)	